

Chapter 61: Digital Learning and FOMO among Adolescents: Challenges to Mental Well-Being

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Abstract

Digital learning has transitioned from a secondary instructional approach into a fundamental component of contemporary education, offering adolescents greater flexibility in learning. Fear of missing out (FOMO) is a common psychological problem among adolescents. It is characterized by persistent worry that an individual is being excluded from valuable opportunities, crucial information, or shared experiences enjoyed by others. While technology and internet access provide adolescents with vast opportunities for digital learning, FOMO simultaneously leads them to shift their focus from academic tasks to digital notifications or digital social interactions. This chapter highlights the challenges of digital learning and fear of missing out among adolescents. During the development of online learning paths, schools must consider and mitigate associated risk factors, and need to maintain a balanced and supportive digital environment. Digital learning and FOMO combine to create a negative feedback loop. This loop leads to constant connectivity anxiety, sleep disturbance, social comparison anxiety, low self – esteem and academic stress, digital access gap, headaches and fatigue. These factors collectively impact adolescents overall mental well – being.

Keywords: Digital Learning, Fear of Missing Out (FOMO), Adolescents, Mental Well - Being, Negative Feedback Loop, Academic Stress, Social Comparison, Anxiety.

1. Introduction

In contemporary education digital learning plays a central role it helps students to learn outside traditional classroom settings. It facilitates personalized learning. The COVID -19 Pandemic signified a turning point, positioning digital learning as a necessity instead of a choice. Digital platforms convert static lessons into interactive ones by using multimedia. The national educational policy 2020 envisions technology – enabled learning as central to transforming India’s teaching learning ecosystem. It emphasizes digital platforms immersive technologies and data driven approaches to ensure access, equity, quality and competency – based education across all stages of schooling (Ministry of education, 2020).

Digital learning and FOMO (Fear of missing out), are closely linked to each other. Digital learning can unintentionally cause FOMO in adolescents. Driven by the fear of missing out on peer interactions and school updates or educational updates, learners experiencing with high FOMO constantly track digital platforms which harms their mental health by creating anxiety, low self - esteem and disrupted sleep patterns. Fear of Missing out (FOMO) is defined as an “a form of social anxiety marked by persistent belief that others are experiencing rewarding events in one’s absence” Przybylski et al., (2013).

2. Review of Related Literature

In a study by Alzahrani et al. (2026) on university students, FOMO was linked to problematic social media use and poorer sleep quality. Problematic social media use, in turn was linked to lower academic

productivity. These results emphasize the need for promoting healthy digital habits to improve student well – being and academic performance. Similarly a study by Giancola et.al. (2025) titled “Adolescence and online vulnerability: The role fear of missing out (FOMO): A cross – sectional study during the third wave of the COVID -19 pandemic” found that FOMO mediated the link between age and online vulnerability. Younger adolescents were more vulnerable due to higher FOMO. These findings highlight the need to address emotional factors to reduce online risks’ carrying implications for internet literacy and prevention along with limitations and future research are discussed. Amran et al. (2025) examined adolescents’ FOMO during public health emergencies in Malaysia. Findings show early signs of digital mental health issues. Promoting positive online use and raising awareness of digital risks is important. Better health communication is needed to help adolescents understand digital well – being and adapt to new norms. Masri-Zada et al. (2025) reported that mental health problems among adolescents, such as anxiety, depression, ADHD, and body image issues, have grown in recent years. Social media affects brain development and behavior. While it allows connection, it also brings risks such as addictive use and exposure to unrealistic standards. The study by Przybylski et al. (2013) “Motivational, emotional and behavioral correlates of fear of missing out ” first defined FOMO as “ a pervasive apprehension that others might be having rewarding experiences from which one is absent”. In their research, Padilla and Arriaga (2025), studied 289 university students, measuring FOMO, psychological well-being and ICT use. Their findings suggest that the constant need to stay online and connected negatively impacts of students’ mental health. The researchers also highlighted the importance of using ICT in education and inclusion in a responsible way. They claimed that with the right strategies, technology can support both academic success and students’ overall well-being instead of harming it. Bano and Bano (2025) conducted a theoretical study on “FOMO (Fear of Missing Out) in the digital age: Analyzing the psychological consequences of persistent Connectivity and online social comparison in adolescents” they found that constant connectivity and online social comparison worsen FOMO levels among adolescents .This study reported that FOMO is associated with increased anxiety, depression, low self-esteem, poor academic performance and disrupted sleep habits. They claimed that adolescents need to support to cope with FOMO in the digital age. Interventions like mindfulness practices, encouraging offline activities.

3. Case Studies: FOMO in digital learning - Challenges to mental well - being among adolescents

3.1 Sampling: A purposive sampling method was used. Five adolescents aged 15-17 were selected from both rural and urban areas of Darrang district of Assam .

3.2 Data collection methods

This chapter is based on the qualitative case study method. Data were collected through semi- structured interviews and observation of 5 adolescents aged 15-17 years. The participants were selected from both rural- urban areas to capture diverse experiences of digital learning. Open - ended questions were used to explore experiences related to anxiety, comparison, distraction and burn out. Names are changed to maintain confidentiality. Participation was voluntary.

3.3 Case study analysis

Case study 1: Compulsive digital learner among adolescents.

Participant profile - Student A, Female, Age -16.

Background - Student A attends 4 hours of online classes daily and depends on 3 Whats App groups for academic updates and study material

Observed behavior-FOMO in digital leaning: She checks her phone every 20-25 minutes due to fear of missing an important PDF, assignment or notice from the group.

Identified Challenges:

1. Fear of Missing Out (FOMO).
2. Constant connectivity anxiety.
3. Sleep disturbance and irregular sleep cycle.

Analysis: The constant need to stay updated creates mental fatigue. This leads to reduced concentration during classes and affects well-being of students.

Implication - Educators and parents need to digital hygiene, like fixed times for checking study groups to reduced anxiety among digital learner.

Case study 2: Social comparison in online platforms.

Participant profile – Student B, Female, age -16.

Background - Student B's school shares assignment marks, quiz scores and submission status publicly on Google classroom.

Observed behavior - FOMO in digital learning - She compares her marks and submission time with classmate and feel anxious if isn't in the top. She spends extra 2-3 hours daily check just to submit before others.

Identified challenges:

1. Social Comparison anxiety.
2. Low self esteem and academic stress.
3. Fear of falling behind.

Analysis: Public academic comparison online increase stress and FOMO. Schools must create safe digital spaces focused on learning not ranking.

Implication: Teacher should focus on individual progress feedback to reduce comparison and build healthy learning environment.

Case study 3: Digital distraction among adolescents

Participant profile: Student C, Male, age -17.

Background: Student C attends 3 hours of online classes on laptop and mobile. During lecture he keeps, Instagram, WhatsApp, Facebook open simultaneously.

Observed behavior - FOMO in digital learning: he fears missing social updates, message while studying. So he switches tabs every 10-15 minutes during class and takes notes irregularly.

Identified challenges:

1. Media multitasking.
2. Poor concentration and attention span.
3. Low retention and declining academic grades.

Analysis: Constant distraction due to fear of missing social content reduces learning effectiveness. Information is not processed deeply which causes learning loss and poor exam performance.

Implication: Educators and parents should promote digital well-being. Fixed study hours to prevent screen time.

Case study 4: Digital divide and access anxiety.

Participant profile: Student D, Male, age 15

Background: Student D is from rural area. He attends 2 hours online class with poor network and one shared device.

Observed behavior- FOMO in Digital learning: He misses live classes due to network device issues. Later he feels anxious that others understand concept he didn't.

Identified challenge:

1. Digital access gap.
2. Academic anxiety and FOMO.
3. Learning gap due to missed classes.

Analysis: Lack of access internet and devices creates gaps in learning. This intensifies fear of falling behind peers, leading to increased academic stress.

Implication: School should ensure equitable digital access and provide record class notes reduce learning gaps.

Case study 5- : Online exhaustion

Participant profile: Student E, female, age -17

Background: Student E spends 9 hours daily on screen

Observed behavior FOMO in digital learning: She joins every extra online session fearing she will miss an important exam trick.

Identified challenge:

Headaches and fatigue

Analysis: Too much screen time reduces focus and retention and causes headache and fatigue.

Implication: Educators and parents should promote digital well-being, fixed study hours to prevent screen time.

4. Limitations of the study

This case study was conducted with five adolescents aged 15-17 from rural and urban areas of Darrang, Assam. Using purposive sampling. Due to the small sample size, specific location, and limited age group, the results can't be generalized. Further studies with larger samples across different regions are recommended.

5. Conclusions

In conclusion, this study highlights that FOMO is a major concern in digital learning among adolescents. Due to fear of missing updates, comparison, distractions, and unequal access, adolescents face anxiety, poor concentration, sleep disturbance, and academic stress. To reduce FOMO and its effects in digital learning. Schools' and parents should encourage healthy digital habits, equal access, and focus on personal growth over competition. A balanced and supportive digital environment is essential to protect adolescents' mental well-being and ensure effective learning.

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